Question	Answer	Marks	Guidance
8(a)	packet / quantum of energy	M1	**Allow** 'discrete amount of energy'. (Not just 'value' for 'amount'). **Allow** 'quanta' or 'bundle' for 'packet'.
	of electromagnetic radiation	A1	**Allow** 'em' for 'electromagnetic'. **Allow** 'wave' or 'emission' for 'radiation'. **Ignore** any specific named part of the spectrum (such as 'light'). **Do not allow** the second mark if the idea is conveyed that the em radiation is 'carrying' the photons. (It needs to be clear that the photons are the em radiation). E.g. not 'in' or 'from' for 'of'. 'Quantum / packet of electromagnetic energy' is awarded full credit.
8(b)(i)	electrons	B1	
8(b)(ii)	electrons are decelerated by target	B1	**Allow** 'charged particles' for 'electrons' throughout (but **not** an incorrectly named particle). **Allow** 'anode' for 'target' **Allow** 'metal' for 'target'
	(some) kinetic energy of electrons becomes energy of photons	B1	**Allow** 'X-ray' for 'energy of'

Question	Answer	Marks	Guidance
8(b)(iii)	some electron kinetic energy dissipated as thermal energy (in target)	B1	**Do not allow** 'electrons have a range of kinetic energies' **Allow** 'heat' for 'thermal energy'. **Allow** omission of 'electron' or 'kinetic' but not both. **Allow** 'electron KE' for 'electron kinetic energy'.
	wavelength (of X-rays) is inversely proportional to photon energy	B1	**Ignore** $E = hc / \lambda$ unless E defined as photon energy, λ defined as wavelength, and hc stated as constant
	maximum photon energy (and therefore minimum wavelength) when all of the electron kinetic energy becomes the energy of a photon	B1	**Allow** omission of 'electron' or 'kinetic' but not both. **Allow** 'electron KE' for 'electron kinetic energy'.
8(b)(iv)	$hc / \lambda = qV$	C1	**Allow** $hf = qV$ **and** $f = c / \lambda$ **Allow** e for q. No credit for $hv / \lambda = qV$ unless v is clarified as speed of light in substitution.
	$\lambda = (6.63 \times 10^{-34} \times 3.00 \times 10^{8}) / (1.60 \times 10^{-19} \times 52.0 \times 10^{3})$	C1	
	$= 2.39 \times 10^{-11}$ m	A1	Correct to at least 3 significant figures. AFC. (2.391)
8(c)	discussion of differences in (linear) attenuation coefficient	B1	**Allow** 'absorption coefficient' for 'attenuation coefficient'
	good contrast when the transmitted intensities (through the two materials) are very different	B1	**Allow** 'not good contrast when the transmitted intensities (through the two materials) are similar'. **Allow** 'pass through' for 'transmitted'